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**Roll No.** | | | | | | | | | | | | | **Total No. of Pages : 02**

**Total No. of Questions : 18**

**B.Tech. (CE) (2012 to 2017) (Sem.–4)**

# CONSTRUCTION MACHINERY AND WORKS MANAGEMENT

**Subject Code : BTCE-402**

**M.Code : 56084**

**Time : 3 Hrs.** **Max. Marks : 60**

**INSTRUCTIONS TO CANDIDATES :**

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

## SECTION-A

**Answer briefly :**

1. How does Milestone Chart differ from a Bar Chart?
2. Differentiate between Activity and Event.
3. Explain the process of compaction of concrete.
4. What is meant by Direct and Indirect costs of a project?
5. List two advantages of two wheel Type Tractors.
6. Discuss Fulkerson's rules for numbering the events of a network.
7. How will you draw the cost duration curve?
8. Draw typical cost duration curve.
9. Define the term Line of Balance.
10. What does ITS and FS stand for?

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## SECTION-B

11. Explain the factors affecting the Economic Life of a construction equipment.
12. Explain the functioning of Dragline in detail alongwith line diagram.
13. How batching of concrete is done for concrete. Explain.
14. What is the effect of grade on cost of hauling earth with trucks?
15. How placing of concrete in cold weather is done?



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## SECTION-C

16. From the data given in the table for a CPM project, draw the network and determine the critical path based on total float.

| Activity | Immediate Successor | Duration (days) |
|---|---|---|
| A | C | 3 |
| B | D,E | 5 |
| C | F,G,H | 4 |
| D | F,G,H | 8 |
| E | I | 9 |
| F | J,K,L,M | 5 |
| G | K,L,M | 8 |
| H | I | 6 |
| I | P | 5 |
| J | N | 4 |
| K | O | 7 |
| L | Q | 6 |
| M | P | 7 |
| N | R | 4 |
| O | R | 8 |
| P | R | 4 |
| Q | R | 5 |
| R | None | 3 |

17. A maintenance project consists of number of jobs. Their normal duration and costs alongwith crash costs and duration are given below. Find out the optimum project cost and time

| Job | Normal duration | Normal cost (Rs) | Crash duration | Crash cost (Rs.) |
|---|---|---|---|---|
| **1-2** | 9 | 9000 | 6 | 15000 |
| **1-2** | 8 | 2000 | 3 | 9500 |
| **1-4** | 15 | 5000 | 10 | 2000 |
| **2-4** | 5 | 2000 | 3 | 4000 |
| **3-4** | 10 | 7000 | 6 | 13000 |
| **4-5** | 2 | 3000 | 1 | 7000 |

Indirect costs are Rs. 6000/- per day.

18. Write a short note on the following :

a) Time grid diagram

b) Tractors

c) Bitumen Plants

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